

BASIC EQUATIONS FOR ME 200

Mass Conservation, 1st Law and 2nd Law Relations

$$\left. \frac{dm}{dt} \right|_{\text{system}} = \sum_{\text{in}} \dot{m}_{\text{in}} - \sum_{\text{out}} \dot{m}_{\text{out}}$$

$$\left. \frac{dE}{dt} \right|_{\text{system}} = \sum_{\text{in}} \dot{m}_{\text{in}} (h + ke + pe)_{\text{in}} - \sum_{\text{out}} \dot{m}_{\text{out}} (h + ke + pe)_{\text{out}} + \dot{Q} - \dot{W}$$

$$\left. \frac{dS}{dt} \right|_{\text{system}} = \sum_{\text{in}} \dot{m}_{\text{in}} s_{\text{in}} - \sum_{\text{out}} \dot{m}_{\text{out}} s_{\text{out}} + \sum_j \frac{\dot{Q}_j}{T_{j,\text{boundary}}} + \dot{\sigma}_{\text{generation}}$$

Work and Heat Transfer Relations

$$W_{\text{boundary}} = \int p \, dV \quad W_{\text{electrical}} = \int \varepsilon \, i \, dt \quad W_{\text{spring}} = \int k_{\text{spring}} x \, dx \quad W_{\text{rot}} = \int \tau \, d\theta$$

$$w_{\text{int,rev,flow}} = - \int_1^2 v \, dp + (V_1^2 - V_2^2)/2 + g(z_1 - z_2) \quad q_{\text{int,rev}} = \int T \, ds$$

PROPERTY RELATIONS FOR ME 200

General Relations

$$h = u + pv \quad c_v = \left(\frac{\partial u}{\partial T} \right)_v \quad c_p = \left(\frac{\partial h}{\partial T} \right)_p \quad E = m(u + ke + pe), ke = V^2/2, pe = gZ$$

$$Tds = du + p \, dv \quad Tds = dh - v \, dp \quad z_{\text{mixture}} = x z_g + (1 - x) z_f, z = \text{property}$$

Models

$$pV = mRT$$

$$R = \frac{\bar{R}}{M}$$

$$R = c_p - c_v$$

$$k = \frac{c_p}{c_v}$$

$$du = c_v dT \quad dh = c_p dT \quad ds = c_v \frac{dT}{T} + R \frac{dv}{v} \quad ds = c_p \frac{dT}{T} - R \frac{dp}{p} \quad s_2 - s_1 = s_2^o - s_1^o - R \ln \left(\frac{p_2}{p_1} \right)$$

$$\left(\frac{p_2}{p_1} \right)_s = \left(\frac{p_{r2}}{p_{r1}} \right)_s$$

$$\left(\frac{v_2}{v_1} \right)_s = \left(\frac{v_{r2}}{v_{r1}} \right)_s$$

$$\left(\frac{T_2}{T_1} \right)_s = \left(\frac{p_2}{p_1} \right)_s^{\frac{k-1}{k}}$$

$$\left(\frac{T_2}{T_1} \right)_s = \left(\frac{v_1}{v_2} \right)_s^{k-1}$$

$$du = c \, dT$$

$$dh = c \, dT + v \, dp$$

$$ds = c \frac{dT}{T}$$

$$h_{\text{comp liq}}(p, T) = h_f(T) + v_f(T)[p - p_{\text{sat}}(T)]$$

EFFICIENCY AND CYCLE RELATIONS FOR ME 200

$$\eta_{\text{turbine}} = \frac{w_{\text{act}}}{w_s}$$

$$\eta_{\text{compressor}} = \eta_{\text{pump}} = \frac{w_s}{w_{\text{act}}}$$

$$\eta_{\text{HX}} = \frac{(h_{H,\text{in}} - h_{H,\text{out}})}{(h_{H,\text{in}} - h_{C,\text{in}})}$$

$$\eta_{\text{nozzle}} = \frac{V_{2,\text{act}}^2}{V_{2,s}^2}$$

$$\eta_{\text{th}} = \left| \frac{W_{\text{net}}}{Q_{\text{hot}}} \right|$$

$$\text{COP}_R = \left| \frac{Q_{\text{cool}}}{W_{\text{net}}} \right|$$

$$\text{COP}_{\text{HP}} = \left| \frac{Q_{\text{hot}}}{W_{\text{net}}} \right|$$

$$\text{mep} = \frac{w_{\text{net}}}{v_{\text{max}} - v_{\text{min}}} \quad \text{bwr} = \frac{|w_c|}{w_t}$$

$$\eta_{\text{th,rev}} = 1 - \frac{T_C}{T_H}$$

$$\text{COP}_{R,\text{rev}} = \frac{T_C}{T_H - T_C}$$

$$\text{COP}_{\text{HP,rev}} = \frac{T_H}{T_H - T_C}$$

$$\left(\frac{T_C}{T_H} \right)_{\text{rev}} = \left(\frac{Q_C}{Q_H} \right)_{\text{rev}}$$

UNIT CONVERSION RELATIONS

Density	1 g/cm ³ = 10 ³ kg/m ³
Length	10 ² cm = 1 m
Velocity	1 km/h = 0.621 mile/h
Volume	10 ⁶ cm ³ = 1 m ³ 1 L = 10 ⁻³ m ³
Force	1 N = 1 kg-m/s ²
Pressure	1 Pa = 1 N/m ² 1 bar = 10 ⁵ N/m ²
Energy	1 J = 1 N-m

Energy Transfer Rate	1 W = 1 J/s 1 kW = 1.34 hp
Specific Heat	1 kJ/(kg-K) = 0.239 kcal/(kg-K)
Universal Gas Constant	$\bar{R}$ = 8.314 kJ/(kmol-K)
Standard Gravity Acceleration	g = 9.81 m/s ²
Standard Atmospheric Pressure	1 atm = 1.01 bar
Temperature Relations	$T(^{\circ}\text{R}) = 1.8 T(\text{K})$ $T(^{\circ}\text{C}) = T(\text{K}) - 273.15$ $T(^{\circ}\text{F}) = T(^{\circ}\text{R}) - 459.67$